FLATNESS OF A MODULE MODULO THE INTERSECTION OF TWO IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Flatness tested through the injectivity of the multiplication map J tensor M to M for every ideal J. the author cites Algebra, Lemma algebra-lemma-flat at proof line 3961. Supplies the criterion that the proof verifies at the end.
- Elementary properties of the tensor product with a quotient, identifying J/(J cap I_1) tensor M with (J + I_1)/I_1 tensor over R/I_1 with M/I_1 M. used at proof lines 3963-3970. Produces the first injection from the flatness hypothesis on M/I_1 M.
- The commutative diagram comparing J tensor M, M and M/I_1 M, and the reduction of the inner term to the second quotient. used at proof lines 3971-3989. Combines the two flatness hypotheses and yields injectivity.

Source reference. Original lemma 0522: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6555–6555. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 639–639. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be an R-module. Let I_1, I_2 be ideals of R. If M/I_1M is flat over R/I_1 and M/I_2M is flat over R/I_2, then $M/(I_1 \cap I_2)M$ is flat over $R/(I_1 \cap I_2)$.*

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Proof. By replacing R with $R/(I_1 \cap I_2)$ and M by $M/(I_1 \cap I_2)M$ we may assume that $I_1 \cap I_2 = 0$. Let $J \subset R$ be an ideal. To prove that M is flat over R we have to show that $J \otimes_R M \to M$ is injective, see Algebra, Lemma 00HD. By flatness of M/I_1M over R/I_1 the map

$$J/(J \cap I_1) \otimes_R M = (J + I_1)/I_1 \otimes_{R/I_1} M/I_1M \longrightarrow M/I_1M$$

is injective. As $0 \to (J \cap I_1) \to J \to J/(J \cap I_1) \to 0$ is exact we obtain a diagram

$$\begin{array}{ccccccc} (J \cap I_1) \otimes_R M & \longrightarrow & J \otimes_R M & \longrightarrow & J/(J \cap I_1) \otimes_R M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ M & = & M & \longrightarrow & M/I_1M & & \end{array}$$

hence it suffices to show that $(J \cap I_1) \otimes_R M \to M$ is injective. Since $I_1 \cap I_2 = 0$ the ideal $J \cap I_1$ maps isomorphically to an ideal $J' \subset R/I_2$ and we see that $(J \cap I_1) \otimes_R M = J' \otimes_{R/I_2} M/I_2M$. By flatness of M/I_2M over R/I_2 the map $J' \otimes_{R/I_2} M/I_2M \to M/I_2M$ is injective, which clearly implies that $(J \cap I_1) \otimes_R M \to M$ is injective. $\square$